

Omnivac Manual

July 14, 2025

1 Sample Stage Rotator

1.1 Sidebar

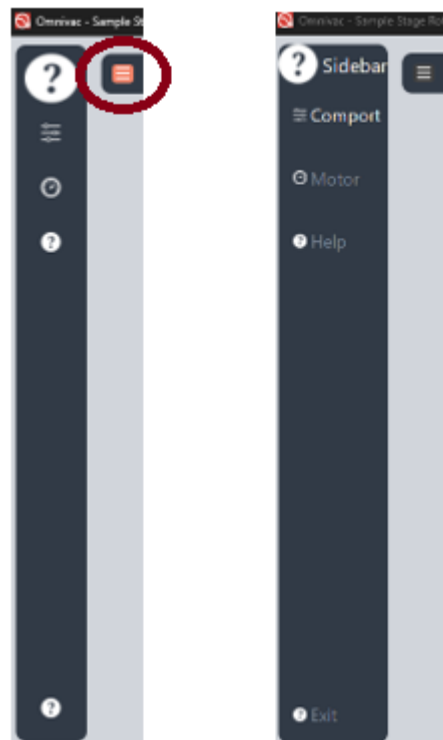


Figure 1: Sidebar

To extend or retrieve the Sidebar you need to click on the button marked by a red circle in Figure 1.

1.2 Comport Page



The image shows a software interface for configuring a serial connection. At the top, there are five dropdown menus: the first is set to 'COM5', the second to '9600', the third to '8', the fourth to 'None', and the fifth to '1'. Below these is a dark button labeled 'Connecte'. At the bottom is a dropdown menu labeled 'Motor7'.

Figure 2: Comport Page

1.2.1 Port

With the first widget, you select the port you want to connect to. Port names are usually something like "COM5".

1.2.2 Baud rate

For motors installed by Omnivac, the baud rate is typically "9600". If a different motor is installed, please refer to its manual to determine the correct baud rate.

1.2.3 Byte size

For motors installed by Omnivac, the byte size is typically "8". If a different motor is installed, please refer to its manual to determine the correct byte size.

1.2.4 Parity

For motors installed by Omnivac, the parity is typically "None". If a different motor is installed, please refer to its manual to determine the correct parity.

1.2.5 Stop bit

For motors installed by Omnivac, the stop byte is typically "1". If a different motor is installed, please refer to its manual to determine the correct stop byte.

1.2.6 Establishing the Connection

After pressing the Connect button, the program will connect to the selected port. Next, simply select the desired motor connected to that port. Once the motor is selected, you can open the sidebar to access the Motor Page.

1.3 Motor Page



Figure 3: Motor Page

1.3.1 Continuous Rotation

This Button leads to a continuous Rotation of the Motor. The direction is depending on the Device Type (Section 2.2)

1.3.2 Presets

On the combobox in Fig.3 with the text Home, are all Presets listed defined in the .ini file (Section 6.3)

1.3.3 Increase/Decrease Speed

As the name implies the buttons are used to increase/decrease the speed of the Motor each click alters the velocity by 12 RPM

1.3.4 User Input

With the input, you enter your desired number, and the motor rotates 360 degrees plus your input in degrees.

1.3.5 STOP Button

This Button is used to stop the Rotation of the Motor

2 Soft Basic

2.1 Device_Name=Axis

Device Name

2.2 Device_Type=1

There are five Device Types:

1. Linear : Device_Type=0
2. Rotation : Device_Type=1
3. Rotation 360 : Device_Type=2
4. Rotation 360-R : Device_Type=3
5. Rotation 360-L : Device_Type=4

There are no differences at this software version between Linear and Rotation Type. In the case of Rotation 360, the values are displayed only in the range of 0 to 360 degree. In the case of Rotation 360-R and L, rotation clockwise and anticlockwise are allowed, respectively.

2.3 Max_Speed(rpm)=100.01

Maximum Speed is set in RPM. This value is used as a regular speed during user set position change.

2.4 Max_Position=3600.00

Maximum Position allowed by the software. Software controls the position so that values above this are forbidden. When user set 4000 the software will follow only position change to the 3600 as above.

2.5 Min_Position=-3600.00

Minimum Position allowed by the software.

2.6 Position_Factor=1.00

Position factor used to calculate real position on a given Axis. Encoder Pulses to be reach = $0.18 * (\text{New Real Position}) * \text{Position_Factor}$

2.7 Position_Unit=°

Position unit is only used to display the right units of a position at the User Spreadsheet

3 Soft Joys

3.1 Joystick_Axis=1

The typical Game Pad exhibit four Analog Axis:

1. Disable : Joystick_Axis=0
2. Axis-X : Joystick_Axis=1
3. Axis-Y : Joystick_Axis=2
4. Axis-Z : Joystick_Axis=3
5. Axis-R : Joystick_Axis=4



When Joystick_Axis is set to 1, the Position of a given Device Address will be affected by the analog positions of an Axis-X. During Regular operation the speed of position change will be equal to a half value of Maximum Speed till the button 5 is pressed. When button 5 is pressed the speed will be equal to the regular Maximum Speed.

The button 7 is used to switch between Preset Values, which are described in the next sections. When button 7 is pressed and the Axis-X is completely Right or left shifted on the Game Pad, the next or previous preset position is followed, respectively.

3.2 Death_Band=10000

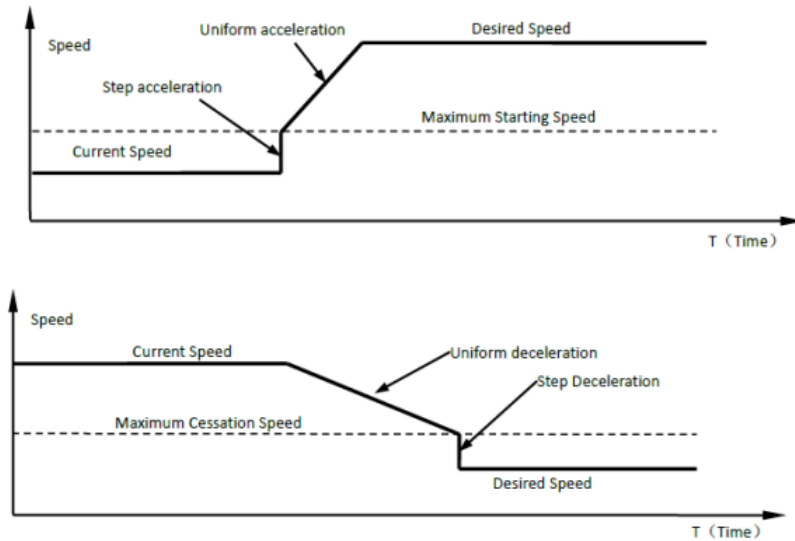
Each Axis of a Joystick exhibit the so called Death Band, the value at which the position is not affected by the joystick.

3.3 Max_Value=65535 Min_Value=0 Center_Value=32768

Each Axis of a Joystick exhibit calibration (Maximum, Minimum and Center) values that can be found in the control panel in the Windows system. Those values must be entered.

4 Soft Moti

UIRobot devices exhibit the Advanced Motion Control Unit that allow for the setting Acceleration and Deceleration Factors according to the figure:



4.1 Acceleration_Rate(ms)=1000

Acceleration Rate is Given in ms.

4.2 Deceleration_Rate(ms)=1000

Deceleration Rate is Given in ms.

4.3 Start_Speed(rpm)=1

Start Speed is given in RPM like regular speed. On the Figure Above it is marked as a Maximum Starting Speed.

4.4 Stop_Speed(rpm)=1

Stop Speed is given in RPM like regular speed. On the figure above it is marked as a Maximum Cessation Speed.

4.5 WARNING

Values described above affect the reaction time of Joystick Operation. When joystick analog Axis is moving back to the 0 position the Advanced Motion Control unit will produce an additional position shift.

4.6 Backlash_Compensation(p)=0

5 Hard Conf

At that section, there are displayed read only values. Those values are not used by the software. They may be used as a information about controller only. For more details please read UIRobot manual.

5.1 Max_Current(A)=2.00

Maximum allowed current by the controller. The current is given in Ampere unit. The allowed values are only two digits (i.e. 2.0).

5.2 Ava_Encoder=1

Encoder Available = 1, Encoder is not present = 0 at the controller

5.3 Ava_CloseLoop=1

Closed Loop Mode Available = 1, Closed Loop Mode not present = 0 0 at the controller

5.4 Ava_AdvMottion=1

Advanced Motion Control Available = 1, Advanced Motion Control Available is not present = 0

5.5 No_IOPorts=4

Number of available IO ports at a given controller

5.6 Soft_Version=1301

Software version

6 Service

At that section, there are displayed read only values. Those values are not used by the software. They may be used as a information about controller only. For more details please read UIRobot manual.

6.1 Pass_Code=0

The typical user can not change the Device Address/Id. The Device Id may be only changed when the right pass code is set at this value

6.2 WARNING

During the Device Address/Id programming the Baud Rate of a RS232-CAN converter is also changed to the 38400 bbps.

6.3 Preset_Position

The software version 0.37 allow to set the so-called preset positions to a given Controller. To set those value one need first to set total number of preset Values. Then each Preset Value is marked in the next line by the Name[X] and Value[X], where X is a order number according to the example bellow. The values are real position values.

```
No=3
Name1=Home
Value1=0
Name2=Gate1
Value2=90
Name3=Gate2
Value3=180
```

6.4 WARNING

The Preset Positions Values must be typed in the rising order !!!! The Preset Positions may be changed by pressing button 7 of a Joystick When button 7 is pressed and the Axis-X is completely Right or left shifted on the Game Pad, the next or previous preset position is followed, respectively.